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Socio-ecological approach to a forest-swamp-savannah mosaic landscape using remote sensing and local knowledge: a case study in the Bas-Ogooué Ramsar site, Gabon

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Socio-ecological approach to a forest-swamp-savannah mosaic landscape using remote sensing and local knowledge: a case study in the Bas-Ogooué Ramsar site, Gabon

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ABSTRACT

Studies of landscape dynamics in protected areas often rely exclusively on remotely-sensed data, leading to bias by neglecting how local inhabitants, who often have a long history of interaction with their environment, perceive and structure the landscape over time. Using a socio-ecological system (SES) approach in a forest-swamp-savannah mosaic within the Bas-Ogooué Ramsar site in Gabon, we assess how human populations participate in landscape dynamics over time. We first conducted a remote sensing analysis to produce a land-cover map representing the biophysical dimension of the SES. This map is based on pixel-oriented classifications, using a 2017 Sentinel-

2 satellite image and 610 GPS points, that categorized the landscape in 11 ecological classes. To study the landscape's social dimension, we collected data on local knowledge to understand how local people perceive and use the landscape. These data were collected through 19 semi-structured individual interviews, three focus groups and three months of participant observation during an immersive field mission. We developed a systemic approach by combining data on biophysical and social dimensions of the landscape. Our analysis shows that in the absence of continued anthropic interventions, both savannahs and swamps dominated by herbaceous vegetation will experience closure by encroaching woody vegetation, leading to eventual biodiversity loss. Our methodology based on an SES approach to landscapes could improve the conservation programs developed by Ramsar site managers. Designing actions at the local scale, rather than applying one set of actions to the entire protected area, allows the integration of human perceptions, practices and expectations, a challenge that is more than essential in the context of global change.

Keywords: landscape dynamics; remote sensing; local ecological knowledge; socio-ecological system; Central Africa

Highlights:

- Integrates remote sensing and local knowledge to develop a socio-ecological approach to the landscape.

- Shows the advantages of using an approach based on local knowledge to map and study tropical wetlands and understand their dynamics.

- Uses comprehension of the links between land use and land cover to produce spatial information on a specific wetland that is difficult to monitor.

- Enables a better understanding of human-environment interactions and their associated ecosystem services within the Bas-Ogooué Ramsar site.

- Can contribute to integration of the perceptions and practices of local populations in the planning of conservation and development programs.

DATA AVAILABILITY STATEMENT

The remote sensing image (2017 Sentinel-2 satellite image) is available online (https://earthexplorer.usgs.gov). Because of restrictions imposed by ethics committee, other data have not been publicly deposited, but our main results can be found on the Vulcar-Fate project website (https://vulcar-fate.obs-mip.fr/). Upon reasonable request, the corresponding author may make available certain data following agreement by the principal investigator of the research project of which this study is a part.

ETHICS DECLARATIONS

Conflict of interest

The authors declare no competing interests.

Consent to participate and publish

All participants in this study were voluntary and signed a consent form after a presentation of the project. During this presentation, the use of the collected data for the publication of scientific papers and reports was explicitly explained. All data collected were anonymized to protect the integrity of the participants.

Ethics approval

All data collection protocols for the project were approved by the France's Institute for Development Research (IRD) Review Board and the VULCARE-FATE ethics committee.

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this study for their hospitality and knowledge, and especially to Rogo, for his daily support all around the landscape mosaic, and to Serge, for his help in organizing focus groups and interviews, in addition to his daily drive of the dugout canoe on the Ogooué River.

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AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Christophe Demichelis and Johan Oszwald. The first draft of the manuscript was written by Christophe Demichelis and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

1. INTRODUCTION

For several decades, remote sensing and analyses using geographic information systems have significantly improved our ability to study landscapes and land-cover changes around the world. These approaches are particularly useful in Central Africa and elsewhere in the intertropical zone where challenging field conditions can hamper on-the-ground access (Laporte et al., 1995; Mayaux et al., 1999; Eva & Lambin, 2000; Kerr & Ostrovsky, 2003; Potapov et al., 2008; Gond et al., 2016; Philippe & Karume, 2019). Thus, diverse applications of remote sensing have been developed to monitor tropical ecosystems. Remote sensing has proven essential to large-scale monitoring of the frequency and intensity of bushfires (Bucini & Lambin, 2002; Jiang et al., 2020), patterns and dynamics of deforestation (Mayaux et al., 2003; Fichet et al., 2012; Kabuanga et al., 2020) and mapping of ecosystem services (Clec'h et al., 2013; De Araujo Barbosa et al., 2015), as well as understanding how diverse anthropic factors affect landscape dynamics (Oszwald et., 2007; Vancutsem et al., 2009; Oszwald et al., 2015).

However, remote sensing alone is insufficient to analyze how local populations perceive, use and transform their environment. The objectives of remote-sensing studies of landscapes often seem restricted to the production of land-cover maps at large scale (Mayaux et al., 2004; Vancutsem et al., 2009; Fichet et al., 2012; Sannier et al., 2014), without taking into account the human and ecological dynamics at work at the local scale. This failure to integrate local dynamics tends to freeze these maps in time and space. These dynamics can be studied by taking into account local knowledge, and more particularly local ecological knowledge. A rich literature in social sciences examines the potential uses of local ecological knowledge in attempts to better understand social-ecological systems and their dynamics (Huntington, 2000; Silvano & Valbo-Jørgensen, 2008; Mistry & Berardi, 2016). Local knowledge is context-specific, making it suitable for uses such as managing conservation and development programs (Wheeler & Root-Bernstein, 2020) or documenting the impacts of climate change (Reyes-García et al., 2020). However, the integration of local knowledge with remotely sensed data remains marginal, and often problematic (Sulieman & Ahmed, 2013; Yiran et al., 2012). However, some recent studies around the world, particularly in the fields of conservation, development and risk management, have successfully

integrated the two types of information (Del Rio et al., 2018; Demichelis et al., 2020; Bonilla-Moheno et al., 2021). This integration is mainly based on the concept of the socio-ecological system (SES; Ostrom, 2009), by considering the landscape dynamics and structure as the results over time of the interactions between human populations and their environment. The SES approach evaluates the physical and ecological landscape through remote sensing, while the social dimensions are highlighted by the local knowledge and local land-use practices (Demichelis et al., 2020).

It is in this perspective, by crossing analyses of remotely sensed data with the collection of local ecological knowledge, that this study focuses on a particular SES in Gabon, characterized by a forest-savanna-swamp mosaic that is difficult to access in the field and difficult to analyze by remote sensing alone because of its landscape heterogeneity. With 89% of its territory covered by forest (Boucka et al., 2021), Gabon is a sparsely populated country compared to its neighbors, and its rural areas have gradually emptied in favor of urban centers (Pottier et al., 2017). Indeed, in order to create protected areas devoid of human presence and anthropic pressure, conservation policies have sought to relocate rural populations, or at least to limit their access to areas declared as protected. Instead of indigenous people exploiting the resources of their environment for subsistence, rural populations in protected areas came to be seen as poachers (Engo, 2014). However, their removal from protected areas has sometimes had negative consequences. For example, in 2006, populations living on the periphery of Gabon's Lopé National Park were denied access to the park and prohibited from any activities related to the exploitation of its natural resources (Moumaneix & Nkombe, 2017). This decision led to disruption of the SES and acceleration of forest expansion to the detriment of the savannahs (Sebag et al., 2016), owing particularly to a disruption in the practice of bush fires previously managed by local populations. Faced with such unintended negative consequences, and with greater awareness of the long-term co-evolution between human populations, their associated practices and the landscapes, the Gabonese government, and more particularly the National Parks Agency (ANPN) created in 2002, seems today to want to reverse this trend.

The example of Lopé shows that studies at the local scale of the interactions between populations and the landscape are essential for improving the efficiency of

conservation and development policies, especially in a context of global changes. While a land-cover map of the Bas-Ogooué Ramsar site in its entirety, produced by the ANPN (Viennois et al., 2017), already exists, it does not take into account local perceptions of the landscape, and the study from which the map resulted did not address the SES dynamics at work within the landscape. Moreover, although this map corresponds to an identity card of the Ramsar site as a whole, it is of little interest for the implementation of conservation rules and policies, as the human and environmental situations differ greatly from one place to another. Indeed, as the largest Ramsar site in Central Africa, it encompasses significant local variability in landscape and human practices. Because of this variability, it is essential to integrate local populations in the decision-making process of environmental management, as they are the ones who are best able to understand, describe and experience the dynamics at work locally on a daily basis. In this perspective, a recent study showed that conservation and development programs are more efficient at local scale with an endogenous governance, rather than exogenously, where decisions are made by national or regional actors who are often disconnected from local realities (Dawson et al., 2021).

Based on a case study of a local SES, this paper proposes an interdisciplinary and replicable methodology that could enable site managers to develop knowledge for the implementation of integrated landscape management at local scale. However, beyond the methodology itself combining remote sensing and local knowledge, this article also aims to produce a human and environmental report on an area of interest to the Ramsar site managers. The article's main objective is therefore to describe and analyze the local dynamics of the SES via two inputs: remote sensing and local knowledge. In the coming years, the implementation of areas that exclude all human activities ("*conservation intégrale*") within this site is planned. Taking into consideration the spatiality and the temporality of human practices, as well as the expectations of local populations concerning development and potential sources of income, will allow better success of this program and better acceptance by local inhabitants. Without such consideration, the conservation actions risk being confronted with the same pitfalls as in the country's other protected areas.

2. MATERIAL AND METHODS

2.1. Study site

This study was conducted in the Bas-Ogooué Ramsar site, Gabon's largest protected area managed by the ANPN, and more precisely in the villages composing the N'Tyatanga *groupement* along the Ogooué River (Figure 1). Note that a "*groupement*" corresponds to a community of villages with the same administrative center, in this case N'Tyatanga. This "*groupement*" comprises four villages: N'Tyatanga, Amendje, Ompomona and Azo II. However, a fifth village, located outside the N'tyatanga *groupement*, was also included in the sample of villages because its portion of exploited land is exactly the same as that of the inhabitants of Azo II. The local landscape, located in the confluence zone of the Ogooué watershed, is composed of a mosaic of tropical rainforest, savannah and swamps. This equatorial ecosystem experiences four seasons: a major dry season from mid-May to mid-September, a major rainy season from mid-September to mid-December, and a short dry season from mid-December to mid-February, followed by a short rainy season from mid-February to mid-May.

With direct access by the Ogooué River, and located less than 30 kilometers away, Lambaréné is the main local market and the administrative center of the study site. However, there are also many exchanges with Port-Gentil, which can be reached in a few hours by river thanks to the daily boat service, and with Libreville, thanks to regular buses and private transporters from Lambaréné. The heterogeneous local ecosystem supports rich biodiversity, including more than 300 fish species, and various endangered mammals, including large populations of African forest elephants (*Loxodonta cyclotis*). The local population is mainly composed of Myéné people, but Fang and Akélé can also be found, especially in Bordeaux village (see Figure 1 for locations of all mentioned places). The average human density in the study site is less than five inhabitants/km^2, implying a very low anthropic pressure and therefore a largely preserved ecosystem, even though numerous ancient agricultural raised fields in the savannahs indicate a very long history of human settlement in this region of Gabon. Locally, people depend largely on fishing, hunting, and agriculture for their livelihoods and economy. However, the latter two activities are slowing down due to the tightening of conservation policies involving seasonal restrictions on and quotas

for hunting, and increasing conflicts with wildlife, especially elephants, for crops. It should be noted that in the past, artisanal logging was the main source of income, but this activity was banned in 2009, leading to a massive rural exodus and a demographic decline in the villages.

As there are no official and clearly defined administrative boundaries to determine the N'Tyatanga "*groupement*", the study site itself, a small area of approximately 370 km^2 (0°45′S 10°01′E; 0°56′S 10°15′E), was defined by the boundaries of village lands, corresponding to the portions of the landscape exploited by local populations of the studied villages for their activities (Figure 1). These boundaries were established following participatory mapping workshops. Note that fishing and hunting are the two activities carried out farthest from the villages and define these limits.

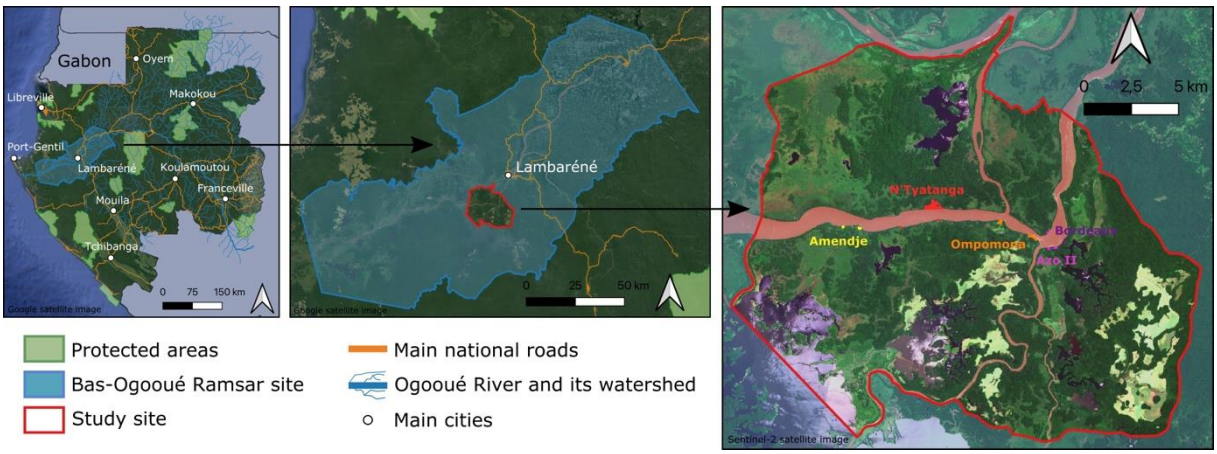

Figure 1: Study site within the Bas-Ogooué Ramsar site.

2.2. Data collection and analyses

To study the landscape through a socio-ecological approach, both qualitative and quantitative data are necessary. These data were collected during an immersive mission of three months between June and September 2022, following a first exploratory mission of 15 days in April 2022. This immersive field mission allowed collection of qualitative and quantitative data about local livelihoods and perceptions of landscapes, through field observations and exchanges with local populations. Geographical and botanical data concerning the structure and dynamics of the

landscape were collected during field visits using a GPS device. All local names collected in the Myéné language are written in [square brackets]. Finally, note that the following subsections are presented in the chronological order of their implementation in the field, highlighting an approach through local knowledge, using ethnographic methods, before conducting analyses of remotely sensed data.

2.2.1. Qualitative data on livelihoods and landscape perceptions

Different qualitative methodologies from the social sciences were used to collect local knowledge about inhabitants' perceptions of the landscape, the exploitation of natural resources, and more generally the livelihoods of local populations. Thus, focus groups, semi-structured interviews and participatory observation constitute the panel of qualitative methods used in this study. All surveys were conducted in French and did not require an interpreter, as the entire population is French-speaking.

First of all, work with three focus groups was conducted, the first comprising the inhabitants of N'Tyatanga and Amendje, the second comprising the inhabitants of Ompomona, and the last the inhabitants of Azo II and Bordeaux. Each of these focus groups included approximately ten people, including men and women of all ages, all engaged in the human activities practiced locally, namely agriculture, fishing, hunting and gathering or logging. In the form of a mapping workshop, the first objective of these focus groups was to delimit the village lands, i.e., the part of the landscape used by the inhabitants of each village for their activities, thus defining the study area. In the second part of work with the focus groups, discussions and debates allowed defining a classification of the landscape shared by all. This led to the identification of the different land units comprising the landscape according to local perceptions. These classes were validated by consensus when they had a name in the Myéné language. The participants were then asked to describe the various human activities related to each land unit.

In a second step, 19 individual interviews were conducted, first with elderly people widely recognized as knowledge holders, and then with younger men and women. These interviews aimed mainly at obtaining information about changes over time in the

landscape and human activities in order to better understand the impact of people on dynamics of the landscape, but no limits were placed on the topics discussed.

Finally, daily participant observations were also conducted to verify the information obtained during focus groups and interviews. The activities carried out by the inhabitants were observed and described.

2.2.2. Quantitative data on land use

After determining, through focus groups, the different classes of land units that comprise the landscape according to local perceptions, a questionnaire was conducted with 66 men and women living in the area. They were asked to name the activities that they personally carried out in each class, and to rate the value of the class for the corresponding activity between 0 and 5. In a second step, they were asked to give two global scores to the land-unit classes, one for the socio-economic value and the other for the socio-cultural value, both scores being also between 0 and 5. This questionnaire allows quantifying the importance of the different classes for human activities and practices, and determining the land uses within the local landscape.

2.2.3. Geographical and botanical data

Two methods were used to collect GPS points related to different ecological stages of land-unit classes. First, after determining the main classes of land units corresponding to local perceptions of the terrestrial landscape, different places were visited with a local assistant. During these visits, these land-unit classes were subdivided into subclasses based on structure and botanical composition corresponding to different ecological states of land unit development that could be used in analyses of remotely sensed data. For these ecological subclasses, multiple criteria were considered: (a) soil water regime (terra firma, floodable or permanent swamp), (b) maximum canopy height, (c) canopy openness, (d) botanical composition (dominant species and species of interest to the population), and (e) potential uses by the human population. In a second step, as a control, the same type of survey was carried out during participant observations. The informant was asked to describe the land-unit classes encountered

during their activities, before taking a GPS point and reclassifying it according to ecological subclass.

All surveys were georeferenced using a GPS device. Each survey point corresponds to a homogeneous space within a radius of 30 m around the GPS point, so that survey points can be used on a Sentinel-2 satellite image with an acceptable margin of error by using 10- and 20-m resolution bands. A total of 610 GPS points were collected and divided into 10 subclasses that can be used for remote sensing. Note that no GPS points were collected for water surfaces, as this eleventh class corresponds to a particular class that can be directly identified and isolated by remote sensing on the satellite image.

Data on floristic composition were first collected in Myéné language according to the knowledge of the assistant, and a photographic database of the flora of interest for each ecological subclass was created. In a second step, species were identified with their scientific names, when possible, based on the first author's personal knowledge from previous experience in Central Africa and consultation of three books on the flora of Gabon (White and Abernethy, 1997; Meunier et al., 2015; Vande Weghe et al., 2016).

2.2.4. Remote sensing analysis

A land cover map was produced by associating GPS points collected in the field, which categorized the landscape into 10 landscape subclasses, in addition to the surface waters. These GPS points, for each of which botanical composition, vegetation structure, and soil water regime were described, were used to conduct a remote sensing analysis through a series of pixel-oriented classifications. The 610 GPS points were used to classify the landscape. Before proceeding with the remote sensing analysis, a PCA was performed using R software with all GPS landmarks, using their pixel values for each band of the satellite image. With the results of the PCA, a Monte Carlo test was performed to control the reliability of the field classification dividing the GPS points into 10 subclasses. The results showed that the classification based on the field observations was highly significant correlated with the pixel values of the satellite image bands (p-value < 0.001***), with no outlier GPS points. Based on this

statistical result, we chose to use all GPS points collected in the field to conduct the remote-sensing analysis, and thus not to perform a map verification process. This choice is further justified by the difficulty of mapping wetlands. Indeed, the removal of points from the sample would have resulted in a loss of pixel information, which is essential to ensure the quality of the mapping of this type of ecosystem.

The analysis was performed on a Sentinel-2 satellite image of April 2, 2017, corresponding to the end of the short rainy season. This allowed taking the water level, in addition to vegetation, into consideration in the analyses. Recent studies agree on the advantages and potential of Sentinel-2 satellite images for remote sensing analyses, especially in tropical areas (Frampton et al., 2013; Verhegghen et al., 2016; Van Passel et al., 2020). While satellite data for this region of Africa are mostly of poor quality due to high atmospheric humidity, cloud cover was almost non-existent in the image we used, which was the most recent image of such high quality. All bands at 10-m and 20-m spatial resolution were selected for the various pixel-oriented classifications, corresponding to visible and infrared light wavelengths. This selection thus accounts for variations in plant biomass and water level with high spatial resolution (Frampton et al., 2013). Note that the 20-m bands were first recalibrated to 10-m using the linear function of the Orfeo ToolBox extension of QGIS before conducting the analyses on ENVI, a specialized remote-sensing software. In addition, in order to reduce noise during the analysis, the landscape was first divided into four major groups that could be differentiated by their canopy structure and biomass: (i) mature forests with a canopy over 15 m, (ii) transitional forest areas with a canopy under 15 m, (iii) areas covered by herbaceous vegetation, and (iv) surface waters (Figure 2). Finally, we attempted to classify the landscape using the *random forest* algorithm. However, because this algorithm is mainly based on temporal variations of pixel values, the use of a single satellite image only allowed differentiation of dry herbaceous savannahs.

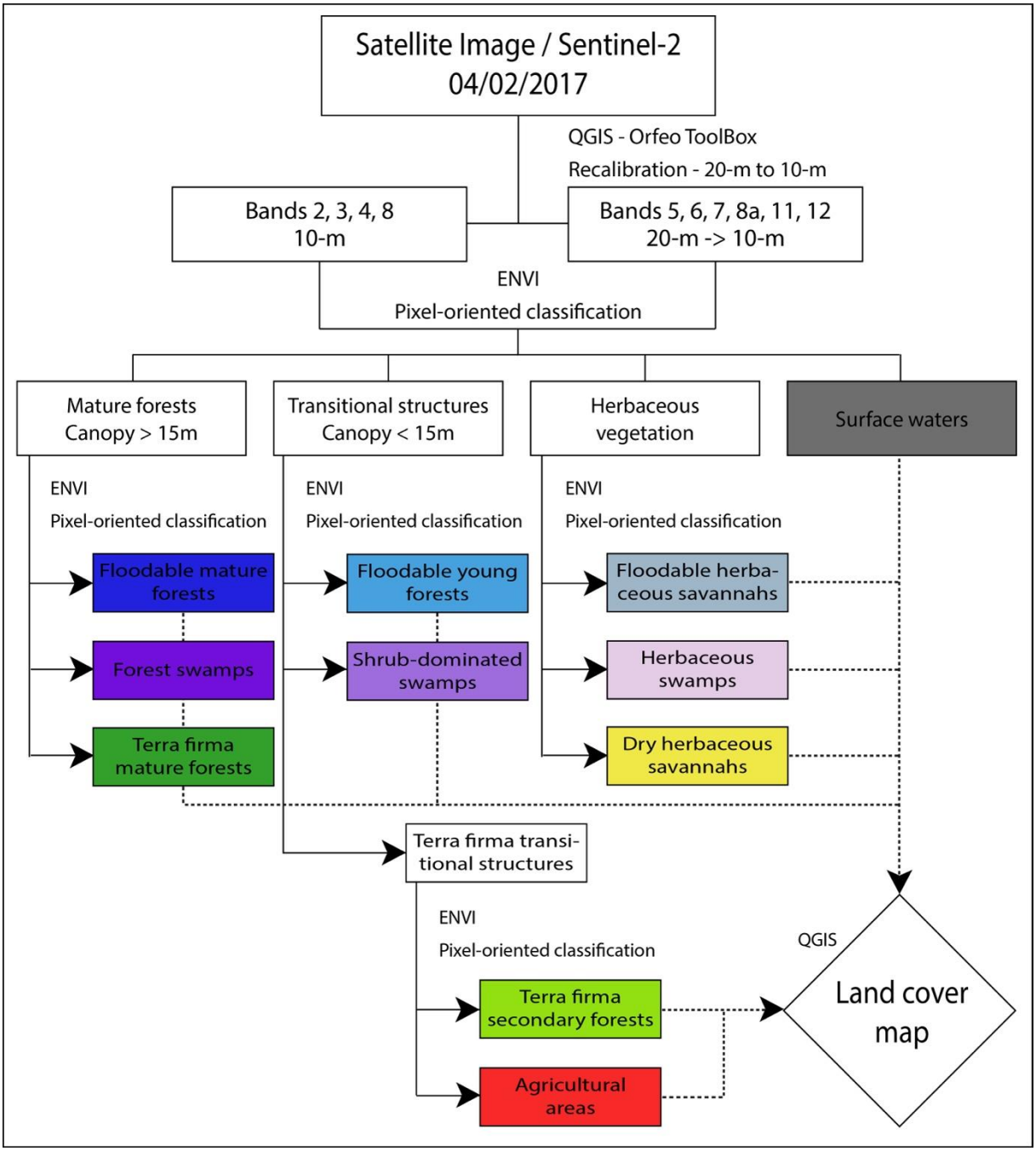

Figure 2: Flowchart of the processing chain used to classify the landscape structure by remote sensing from a Sentinel-2 satellite image.

3. RESULTS

3.1. Landscape composition and dynamics according to remote sensing

3.1.1. Landscape composition according to the land cover map

Based on the processing chain (Figure 2), we produced a land cover map representing the current landscape composition (Figure 3). Each of the subclasses presented in the map is described in the following sections, which detail the ecological dynamics of the landscape. Tables 1 and 2 describe the composition of the landscape according to the different classes defined by remote sensing, as well as more globally according to soil water regimes (Table 1) and vegetation structure (Table 2). The dynamics of the landscape, and more precisely those linked to ecological succession, are largely linked to the soil water regime, which is why the descriptions of the classes in the following parts are organized according to the classification in Table 1.

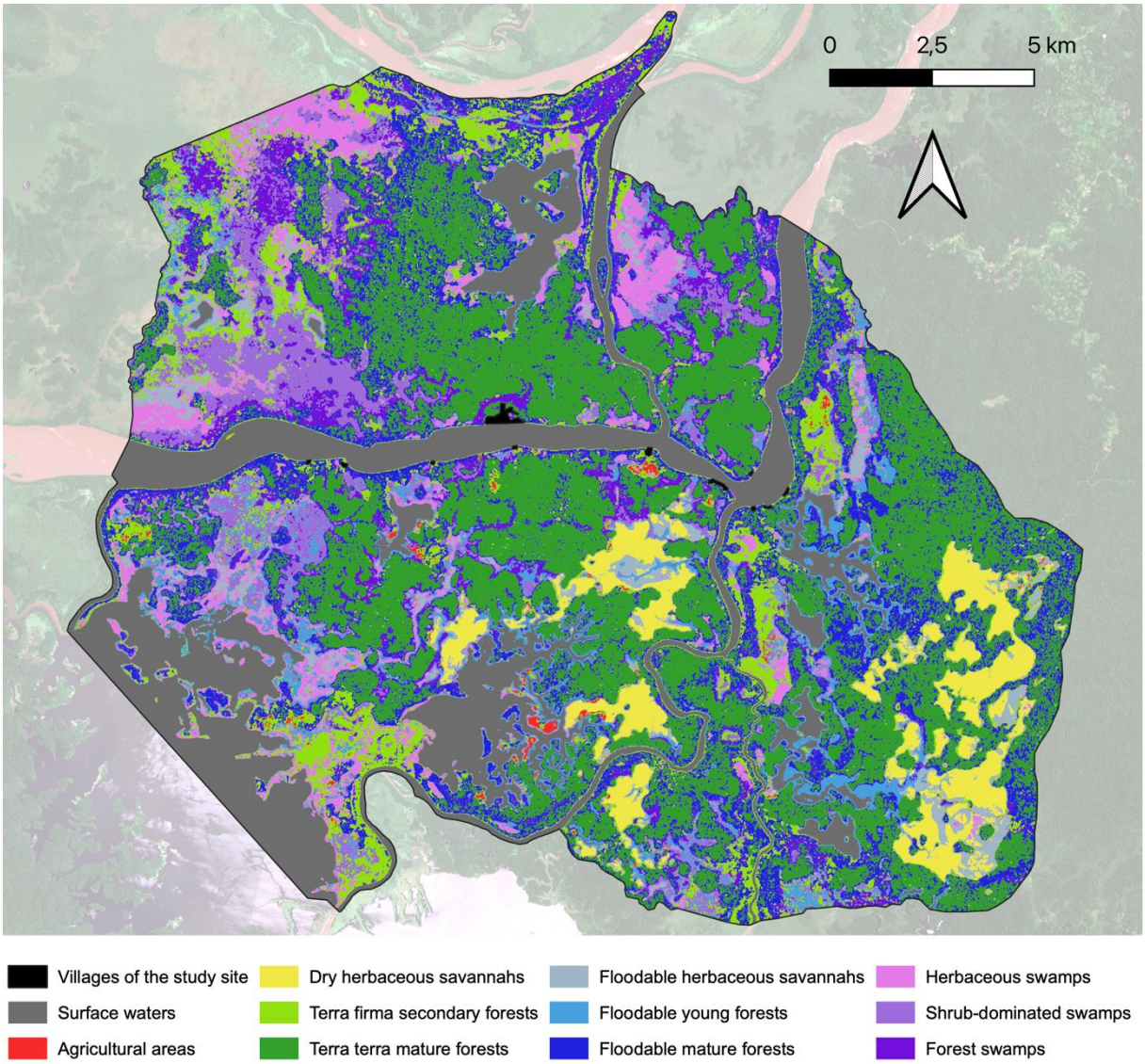

Figure 3: Land cover map of the study site based on a 2017 Sentinel-2 satellite image showing 11 landscape classes, mapped by combining field and pixel-oriented classifications.

Table 1 shows the dominance of well-drained soils over the entire study area, forests and savannahs combined (40.6%). This is followed by seasonally flooded soils (24.8%), and then permanently flooded soils (18.4%). However, in the rainy season, with rising water levels, the floodable areas join the swampy areas to form a single flooded network, which then comprises the majority of the terrestrial landscape (43.2%), in addition to the surface waters, which still represent 15.9% of the local landscape. Indeed, during the rainy season, the flooded areas ensure the connection between the swampy areas and the perennial surface waters, whereas during the dry season, the swamps are isolated from the hydrographic network because of the draining of the floodable areas.

Ignoring the soil water regime, Table 2 shows that mature forest areas dominate (51.4%), followed by transitional structures (20.4%) and then herbaceous structures (12.0%). These values highlight a landscape in mutation, marked by progressive expansion of forest, illustrated by a significant proportion of transitional areas destined to reach a forest stage in the absence of exploitation (agriculture and logging) or fire.

Finally, the agricultural areas, corresponding to the opening of terra firma forests and defining a unique class in both tables, represent a very small proportion of the landscape (0.3%). This value confirms the very low anthropic pressure from agriculture on the forest ecosystems. Furthermore, it is important to note that these agricultural areas are mostly located near surface water, whether along the river, on lakeshores, or on islands.

Tables 1 and 2: Area extents and proportions of the 11 land cover classes for the entire study area, organized by soil water regime (Table 1 left) and vegetation structure (Table 2 right).

Land cover classes (soil water regime)	Area extent (km2)	Landscape proportion (%)
Dry herbaceous savannahs	15,6	4,2
Terra firma secondary forests	31,6	8,6
Terra firma mature forests	102,6	27,8
Total well-drained soils	**149,8**	**40,6**
Floodable herbaceous savannahs	9,5	2,6
Floodable young forests	19,3	5,2
Floodable mature forests	62,7	17,0
Total seasonally flooded soils	**91,5**	**24,8**
Herbaceous swamps	19,1	5,2
Shrub-dominated swamps	24,4	6,6
Forest swamps	24,6	6,7
Total permanently flooded soils	**68,0**	**18,4**
Agricultural areas	**1,2**	**0,3**
Surface waters	**58,6**	**15,9**

Land cover classes (vegetation structure)	Area extent (km2)	Landscape proportion (%)
Herbaceous swamps	19,1	5,2
Floodable herbaceous savannahs	9,5	2,6
Dry herbaceous savannahs	15,6	4,2
Total herbaceous structures	**44,2**	**12,0**
Shrub-dominated swamps	24,4	6,6
Floodable young forests	19,3	5,2
Terra firma secondary forests	31,6	8,6
Total transitional structures	**75,3**	**20,4**
Forest swamps	24,6	6,7
Floodable mature forests	62,7	17,0
Terra firma mature forests	102,6	27,8
Total mature forest structures	**189,8**	**51,4**
Agricultural areas	**1,2**	**0,3**
Surface waters	**58,6**	**15,9**

3.1.2. Dynamics of permanently flooded soils

The herbaceous swamps (5.2%) are mostly located on the edges of lakes, in forest and savannah lowlands, or in vast expanses at elevations close to or below sea level, often near the Ogooué River and its affluents. Vegetation is largely monospecific, with large areas of the sedge *Cyperus papyrus* [nkune]. At the edge of the swamps and at the level of the shallow waters of the lakes it is possible to observe in most cases a species of reed, the grass *Echinochloa stagnina* [oko]. The roots of the papyrus form small loose mounds with emerged tops in the dry season, but are completely submerged in the rainy season. The reeds remain on the surface and follow the variations of the water level. The pioneer tree species *Alstonia congensis* [nguga] can lead to woody encroachment in these herbaceous swamps.

In this second stage of the swamps, the encroachment continues with the death of papyrus and the establishment of several *Raphia* spp. [inimba / ikoto] as well as two pioneer species, *Pseudospondias microcarpa* [osongo] and a species [iniana] whose scientific name could not be determined, as the field surveys took place outside the flowering and fruiting period. Together, these species form shrubby vegetation that can reach 15 m in height at its most advanced stage. This ecological class corresponds to shrub-dominated swamps (6.6%) and the canopy is structured by the oldest individuals of *Alstonia congensis* and *Pseudospondias microcarpa*.

With the drying of the swamps due to the accumulation and sequestration of organic matter, particularly *Raphia* leaves (which were formerly extensively harvested for

construction, but are not exploited today), *Alstonia congensis*, and [iniana] are gradually dying due to lack of water, and only *Pseudospondias microcarpa* and *Raphia* spp. remain. The opening of the environment gradually gives way to larger trees that dominate the canopy and are better adapted to the new soil conditions, with flooding in the rainy season and loose, moist soil in the dry season. These species are sparsely present in association with dense stands of *Raphia* spp. In addition to *Pseudospondias microcarpa*, these large trees include *Treculia africana* [oyaya] and *Guibourtia demeusei* [nkewa]. It is also common to find *Cola acuminata* [ombéni] and *Mitragyna stipul*osa [ntowo] at the periphery of the swamps. Together, these tree species structure the forest swamps (6.7%), whose canopy rarely exceeds 30-40 m. It should be noted that the herbaceous stratum opens up at this stage, allowing easy passage in a dugout canoe during the rainy season, with the frequent presence of *Sclerosperma mannii*, locally called "the straw of the pygmies" [ompavoa akoa] because they use its leaves for their houses. Although the rattan palm *Laccosperma robustum* [nkanda] is not specific to these forest swamps, it is abundantly present.

3.1.3. Dynamics of seasonally flooded soils

The floodable herbaceous savannahs (2.6%) mainly correspond to sedge-dominated vegetation forming a soil composed of clods characteristic of floodable areas. Although numerous sedge species are present, they are all grouped under the name [esinga zorove]. Among them, *Rhynchospora corymbosa* and an unidentified species of *Cyperus* dominate. A woody plant or small shrub with purple flowers, *Melastomastrum segregatum* [nyengele], is also present in these areas. During the rainy season, when the water is at its highest, these areas are totally flooded and the vegetation completely submerged, creating temporary lakes and streams, and therefore waterways that can be navigated by dugout canoe.

The development of young seasonally flooded forests (5.2%) within the lowlands of these savannahs begins with the establishment of *M. segregatum*, in a form that can reach two m in height, and *Anthocleista vogelii* [osanza], leading to the formation of patches of woody vegetation. However, near the rivers and lakes, in areas subjected to longer periods of flooding, the related species *A. liebrechtsiana*, easily recognizable by its divided trunk reminiscent of the stilt roots of *Rhizophora* mangroves, is

responsible for the encroachment. Because of its morphology, which differs from that of *A. vogelii*, this species is set apart, but appears not to have a local name. Subsequently, the young seasonally flooded forests are mostly comprised of a species of *Uapaca* [ntshombi a pupu] in its bushy form, as well as two other bushy shrubs, one without a local name, *Gaertnera paniculata*, and the second *Chrysobalanus icaco* [mpendo], whose fruits, very sweet, are occasionally consumed locally by hunters and fishermen. In the areas closest to permanent water, a shrub with multiple upright trunks [ntambaramba] can be found in large numbers, but it could not be identified because of the absence of flowers or fruits at the time of fieldwork. The canopy of this bushy complex is mostly between five and 15 m. Owing to seasonal flooding, the undergrowth is open all year round, facilitating movement by dugout canoe in the high-water season, and on foot in the dry season.

Over time, these shrubby forests evolve into floodable mature forests (17.0%), which are mostly located on the periphery of lakes and along rivers, but also within gallery forests crossing the savannahs and near the swampy areas, in zones where the soil dries out completely in the dry season, making them particularly popular hunting areas. The undergrowth of these forests is as open as that of young seasonally flooded forests, and movements are as easy as in those sites. The canopy of these forests can reach 30-40 m and is largely dominated by a species of *Uapaca* [ntshombi], which can in some places form monospecific stands. However, in the direct vicinity of the rivers it is common to find *Ceiba pentandra* [oguma] in quantity, increasing canopy height to 50 m in some places. Other dominant species in these areas include *Daniellia klainei* [olenge / esuna], *Sacoglottis gabonensis* [ozuga], whose fruit is consumed locally by hunters, *Spathodea campanulata* [ntsogo], *Mitragyna stipulosa* [ntowo] and *Irvingia gabonensis* [oba], whose seeds ('almonds') are particularly coveted for the preparation of [odika] bread. In addition, it is important to note that according to local knowledge, these forests can progressively colonize and replace forest swamps over time, notably as a result of the accumulation of organic matter in the swamps that is not evacuated to the river and to the formation of a soil that dries out completely during the dry season.

3.1.4. Dynamics of well-drained soils

The dry herbaceous savannahs (4.2%) are mainly grass-dominated systems. As for seasonally flooded herbaceous savannahs, all the species present are grouped under the single name [esinga zorove]. *Anadelphia afzeliana* and *Hyparrhenia diplandra* are found in abundance, and *Ctenium newtonii*, *Trichanthecium nervatum*, *Imperata cylindrica* and *Pennisetum polystachion* are more scattered. Owing to the biannual practice of setting fires, savannah shrubs are not very abundant, with scattered individuals of only two species, both without local names, *Nauclea latifolia* and to a lesser extent *Crossopteryx febrifuga*, both of which are known to be resistant to fire.

Two types of forest comprise the terra firma secondary forests (8.6%) on the land cover map. However, because their structures are similar, it was not possible to differentiate them using remote sensing, and both have been grouped under the same name. In areas of dry herbaceous savannahs where fire is less intense, i.e., at the edge of the densest forests, near floodable savannahs, and sparsely near savannah shrubs, woody encroachment may occur with *Anthocleista schweinfurthii*, also named [osanza] like its relatives in seasonally flooded vegetation, coupled with various fern species such as *Dicranopteris linearis* and *Pteridium aquilinum* both named locally [osaza], but also *Selaginella myosurus* [aroyroy] and *Cyclosorus striatus* (no local name). This vegetation then gives way to the installation of pioneer forest species that can develop up to 15 meters in height and structure young terra firma forests. *Pentaclethra macrophylla* [mbala], *P. eetveldeana* [ozengezenge], and *Zanthoxylum heitzii* [nongo / olumi] are the main species, but *Trichoscypha acuminata* [owuga] and *Xylopia aethiopica* [ogana] are also commonly encountered. In even further advanced stages, it is frequent to find a high density of young *Aucoumea klaineana* [okumé]. Concerning the herbaceous and shrubby stratum, the ferns give way to a creeping tree species, *Alchornea cordifolia* [mbogo], which gradually colonizes the savannahs, along with *Aframomum* spp. [ntondo]. Note that these afforestation dynamics are also found after natural treefalls within mature forests.

The second type of secondary terra firma forests corresponds to old agricultural fallows, as well as forest spaces opened up by logging with selective exploitation of *A. klaineana*. Vegetation in these sites progressively develops into forest, classified as secondary forest when their canopy does not exceed 15 m. Several pioneer species are characteristic of forests degraded and disturbed by anthropic activities.

Zanthoxylum heitzii [nongo / olumi], *Musanga cecropioides* [nkombo-Gombo], *Myrianthus arboreus* [mbowa], *Distemonanthus benthamianus* [otangani / ogeminya], *Harungana madagascariensis* [osakadi] and to a lesser extent *Pausinystalia johimbe* [yohimbé] are frequently found in these secondary terra firma forests. In contrast to the areas affected by logging and natural treefalls, the fallow lands also present high abundance of *Elaeis guinensis* [oyila] cultivated for its nuts and for wine production, which remain after the fields have been abandoned. These areas often have a very closed herbaceous and shrub stratum, difficult to penetrate due to the abundance of *Alchornea cordifolia*.

Over time, these two types of secondary forest evolve into mature terra firma forests (27.8%). Locally, they are often found on the edges and tops of small hills. They are dominated in the area by *A. klaineana*, which can form mono-dominant forests in some places. The canopy is between 40-50 me, but can reach 60 m in some places thanks to the presence of *Baillonella toxisperma* [oréré] or *Guibourtia tessmannii* [ékewazengo]. This is the ecological landscape class with the greatest diversity of species. In addition to *A. klaineana*, the most characteristic trees structuring the canopy are *Nauclea diderrichii* [mbilinga], *Testulea gabonensis* [izombé] and *Lophira alata* [nkomba]. Also present are *Annickia affinis* [ogowa], *Funtumia africana* [onembo], *Scorodophloeus zenkeri* [ntsinjakolo], *Pycnanthus angolensis* [ilomba], *Staudtia kamerunensis* [ogowéli] and finally *Coula edulis* [ogula] and *Dacryodes edulis* [ozigo], two species whose fruits are harvested and sold. The undergrowth of these forests is often closed, with several species of Marantaceae, bushy shrubs and numerous lianas, making it difficult to move around outside of elephant trails, which are used by hunters.

All terra firma forests can be used for slash-and-burn agriculture. On the land cover map, the agricultural areas (0.3%) correspond to agricultural fields and young fallow lands, which cannot be differentiated from each other by remote sensing because of their similar vegetation structure. The same species of ferns that colonize the savannahs [osaza] are also early colonizers of former agricultural fields, along with *Alchornea cordifolia*. If the fallow period is not long enough and the area is burned regularly, the fallows evolve into monospecific stands of *Pteridium aquilinum* [osaza], slowing down the establishment and development of pioneer tree species by closing the environment. This type of formation is mostly found in fallows on the periphery of

savannahs, at the forest edge, in areas that have undergone repeated intense fires, as the ferns are resistant to fires and grow back directly after them to form these characteristic spaces in disturbed areas. Similarly, in the case of agriculture on islands in lakes, the almost complete felling of trees slows down, or even blocks, forest regeneration, resulting in monospecific stands of ferns that persist over very long periods.

3.2. Livelihood and landscape perceptions

3.2.1. Local perceptions of the landscape

Following the focus groups conducted in each village, a first shared classification of the landscape was identified. This classification divides the landscape into four major land units: (i) surface waters [Imbora s'aningo], (ii) forests [Iga], (iii) savannahs [Orove] and (iv) swamps [Iwongu]. Another particular class with a mainly cultural role can also be identified, the islands, comprising the forest islands in the savannahs and the islands in the lakes [Inange]. Agricultural fields and fallow lands form two special classes not included in this classification of the landscape, but considered as anthropic places in the same way as villages. Thus, fields are called [N'Tyaga], and fallow lands [Oda], the latter term applying even to old fallows that have already reached a forest stage. Note that the Ogooué River [Ogowe] and its affluents [Orembo] together form the structuring axes of the landscape, and can be separated from other surface waters, such as lakes [Eliwa] and ponds [Edje]. Similarly, the main land unit [Iga] can be divided, according to some people, into two sub-units, the seasonally flooded forests [Iga niamangu] and the terra firma forests [Iga ninomba]. However, as that sub-division was not always mentioned during the focus groups, it was decided to keep the main land unit [Iga] as referring to both forest types in quantifying the structure of the landscape. Photos 2 and 3 give an overview of the landscape mosaic of the study area.

Photos 1 and 2: Lake surrounded by forests and herbaceous swamp vegetation (left); Savannah divided by a shrub-dominated swamp with a forest in the background (right)

Table 3 reclassifies the different classes used in remote sensing within this local landscape classification. Thus, the table presents a hybrid classification of the landscape structure, taking into account both soil water regime (Table 1) and vegetation structure (Table 2). Thus, the areas with permanently flooded soils in Table 1 form the same group in this local classification, swamps [Iwongu – 18.4%], regardless of the structure of the vegetation. The savannahs [Orove – 6.8%] correspond to the herbaceous formations in Table 2, but without the swampy areas. According to this same classification, the transitional areas already belong to the dominant forest landscape [Iga – 58.6%], which comprises all the forests with well-drained and seasonally flooded soils (Table 1), whatever their stage of development, but excludes the forests with permanently flooded soils (Table 1). Note that forest islands [Inange] do not appear in Table 3. Indeed, this particular class is not linked to a type of vegetation or soil regime, and can be composed of several different ecological units, although they will necessarily be forest units. Thus, a forest island [Inange] corresponding to an isolated forest patch in savannah can be considered as a sub-unit of the main land unit [Iga].

Tables 3: Area extents and proportions of the 11 land-cover classes for the entire study area, organized by local classification of the landscape.

Land cover classes (local classification)	Area extent (km2)	Landscape proportion (%)
Floodable herbaceous savannahs	9,5	2,6
Dry herbaceous savannahs	15,6	4,2
Total savannahs [Orove]	**25,1**	**6,8**
Terra firma secondary forests	31,6	8,6
Terra firma mature forests	102,6	27,8
Floodable young forests	19,3	5,2
Floodable mature forests	62,7	17,0
Total forests [Iga]	**216,1**	**58,6**
Herbaceous swamps	19,1	5,2
Shrub-dominated swamps	24,4	6,6
Forest swamps	24,6	6,7
Total swamps [Iwongu]	**68,0**	**18,4**
Agricultural areas [N'Tyaga / Oda]	**1,2**	**0,3**
Surface waters [Imbora s'aningo]	**58,6**	**15,9**

3.2.2. Local livelihoods and land exploitation

First, for the same reason as previously mentioned, it was decided to keep the main land unit [Iga], and not the sub-division, to conduct the questionnaires. Table 4 presents the results of the questionnaires. The scores were between 0 and 5 and correspond to averages. The shaded values correspond to the maximum value for each land unit concerning human activities. The values in bold typeface correspond to the maximum values for each activity and global scores (average, economic and cultural).

Table 4: Results of the questionnaires presenting the average values given by the population to the different main classes of land units according to human activities and practices.

	Surface waters [Imbara s'aningo]	**Swamps** [Iwongu]	**Forests** [Iga]	**Savannahs** [Orove]	**Forest islands** [Inange]
Agriculture score	0,13	0,12	**4,19**	0,48	2,57
Hunting score	0,16	3,07	**4,67**	3,22	1,34
Fishing score	**4,61**	3,04	1,33	1,06	0,06
Gathering /logging score	0,06	1,04	**2,07**	0,13	0,27
Average score	1,24	1,82	**3,07**	1,22	1,06
Economic score	**4,63**	2,28	3,46	1,37	0,76
Cultural score	3,16	1,85	**4,79**	1,99	4,09

Looking at the average values for each activity, it is important to note that forests represent the land unit with the highest score (3.07). Indeed, they seem to be useful for all activities, and have the highest scores for agriculture (4.19), hunting (4.67) and gathering (2.07), fishing being unsurprisingly the activity most linked to surface waters (4.61). However, the economic score is highest for surface waters (4.63), not forests.

Thus, it would seem that fishing is the most important income-generating activity locally, while forests are more the seat of subsistence activities, even if they constitute an important economic place since their score remains the second highest (3.46).

After forests, swamps appear to be the second most used land unit, with the second highest average score (1,82) and the third highest economic score (2,28). Indeed, while in the dry season fishing is almost exclusively practiced in surface waters and for commercial purposes, during the rainy season this activity is mostly practiced in swamps and flooded forests and for subsistence purposes. Similarly, hunting is practiced exclusively in terra firma forests during the rainy season, either with guns or snare providing a source of income outside of the peak fishing season. In contrast, during the dry season, hunting is mainly practiced in the dried-up floodable forests, but also in and around swamps, where wildlife gathers to access the last remaining water points.

Regarding savannahs, they are mainly devoted to hunting (3.22), and to a lesser extent to fishing (1.06). Indeed, at the end of the dry season, after the savannah fires, the population takes advantage of this land unit and the regrowth of grasses that attracts wildlife from the forests into open areas. Similarly, in the rainy season, some inhabitants carry out subsidence fishing in flooded savannahs, hence their low economic value (1,37), with fish taking refuge in the temporary streams and ponds formed by the rains.

Finally, the forest islands have a high average score for agriculture (2.57), ranking second. Indeed, a fringe of the population prefers these places on the lakes for agriculture, in particular because they offer a natural protection against the intrusion of wildlife. The most remarkable thing about this land unit is its high cultural score (4.09), just behind forests (4.79), and ahead of surface waters (3.16). Indeed, certain places in the forests, often on hilltops and on many islands in lakes (and forest islands in savannahs), are home to genies and spirits. Access to these places is highly regulated, and only those who are initiated and have a connection with these spirits can enter. To a lesser extent, all the surface waters are also inhabited by mermaids and aquatic monsters, stories about them being abundant.

3.3. Socio-ecological approach to dynamics of the landscape

Based on the information presented in preceding sections and additional results from focus groups, interviews and participant observations, Table 5 synthesizes the links between the land-cover classes and human activities. These links illustrate the provisioning services provided by the local ecosystem to human populations. However, in this synthesis, some activities have been subdivided, such as hunting and fishing, and two additional activities have been included, namely sand extraction for construction and wild honey harvesting. Although these activities were not mentioned in the questionnaire, they were observed in the field. The blue boxes correspond to activities carried out in the rainy season, the yellow boxes to activities carried out in the dry season, and the green boxes to activities carried out throughout the year regardless of season.

Table 5: Location of activities within the land-cover classes based on field observations, focus groups and interviews. In green: activity conducted throughout the year; in blue: rainy season activity; in yellow: dry season activity.

		Surface waters	Herbaceous swamps	Shrub-dominated swamps	Forest swamps	Dry herbaceous savannahs	Terra firma secondary forests	Terra firma mature forests	Floodable herbaceous savannahs	Floodable young forests	Floodable mature forests	Agricultural areas
Slash and burn agriculture												
Hunting	Gun											
	Snare											
	Spear											
Fishing	Net											
	Hook											
	Machete											
Honey harvesting												
Gathering / logging												
Sand extraction												

By combining this information on the use of the landscape by the population and data on ecological succession across the landscape, we produced a systemic approach to landscape dynamics, linking the different ecological classes identified in a dynamic system. This model allows a better understanding of the local socio-ecological system, which is characterized by landscape dynamics driven by both ecological and anthropic factors.

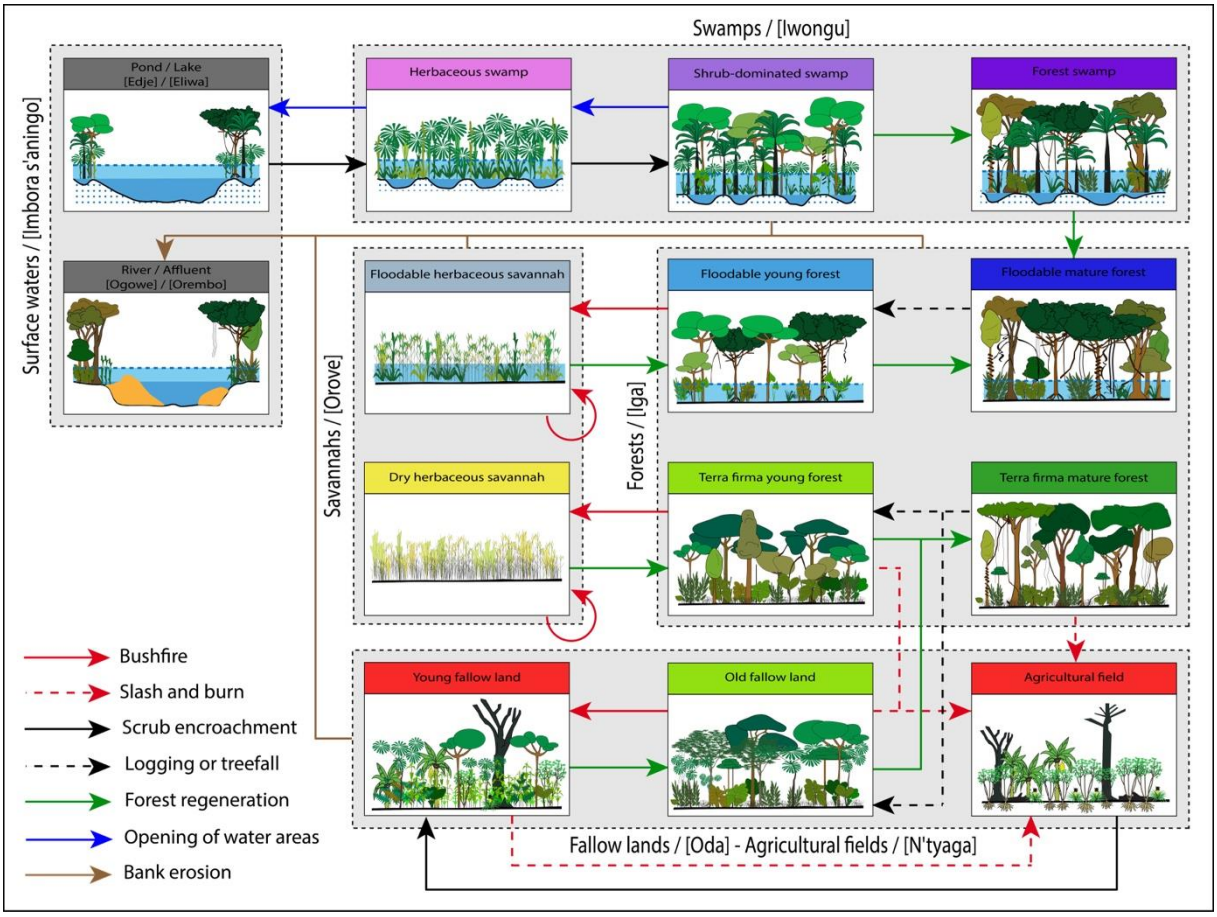

Figure 4: Socio-ecological approach to the landscape dynamics driven by ecological and anthropic factors.

In the absence of anthropic pressure, natural processes tend to close the environment, with a progressive colonization of herbaceous vegetation, whether savannah or swamp, by the forest. This dynamic has been accelerated since the prohibition of artisanal logging of *A. klaineana*, which resulted in local opening of the environment. Despite prohibition of logging, twice-annual burning of savannahs, practiced in particular by hunters, maintains these open formations. In the same way, machete fishing carried out at the end of the dry season in the herbaceous and shrub-dominated swamps near the villages opens these environments and can create temporary ponds. If these small bodies of open water are maintained by people, they can attract aquatic fauna in the rainy season, which are trapped there as the water level recedes, creating dry-season food stocks for the population in case of need. Note that the ecological successional dynamics involve processes working at different time scales. Some processes operate over relatively short time spans, notably the succession of terrestrial

vegetation leading to afforestation of open areas, while others operate over long time frames, especially the sedimentation of ponds and lakes transforming open water into swamps, into shrub-dominated swamps, and finally into permanently flooded forests. It is also important to note the impact of bank erosion in this model. As mentioned by the population, the increase in river traffic, but also the more abundant rains and flooding phenomena due to climate change, lead to significant erosion of the terrestrial landscape as a whole.

A final step in the co-construction of this model of socio-ecological dynamics was its validation, which was, like the rest of the process, a collaborative exercise. A restitution workshop was organized in N'Tyatanga on September 5, 2022, during which the results were presented and then commented on and discussed with the population (Photo 3). The workshop was attended by about 20 people from all villages, who were transported by motorized dugout canoe by the field researcher. In all the villages, the participants were chosen by the populations themselves, with the only condition that there should be as equal a representation of men and women as possible, and that all age groups should be represented. As in the focus groups and interviews, all participants were French-speaking. Overall, all the results presented were validated by the participants, who reached a consensus after discussions led by the older generation, who have a more experienced perspective on the evolution of the landscape than the younger generation. In addition, some land uses and resource exploitation were further explored, particularly in relation to swamps, and some species not identified during the field surveys were finally identified in the Myéné language through the presentation of specimens (fruits, leaves, photos).

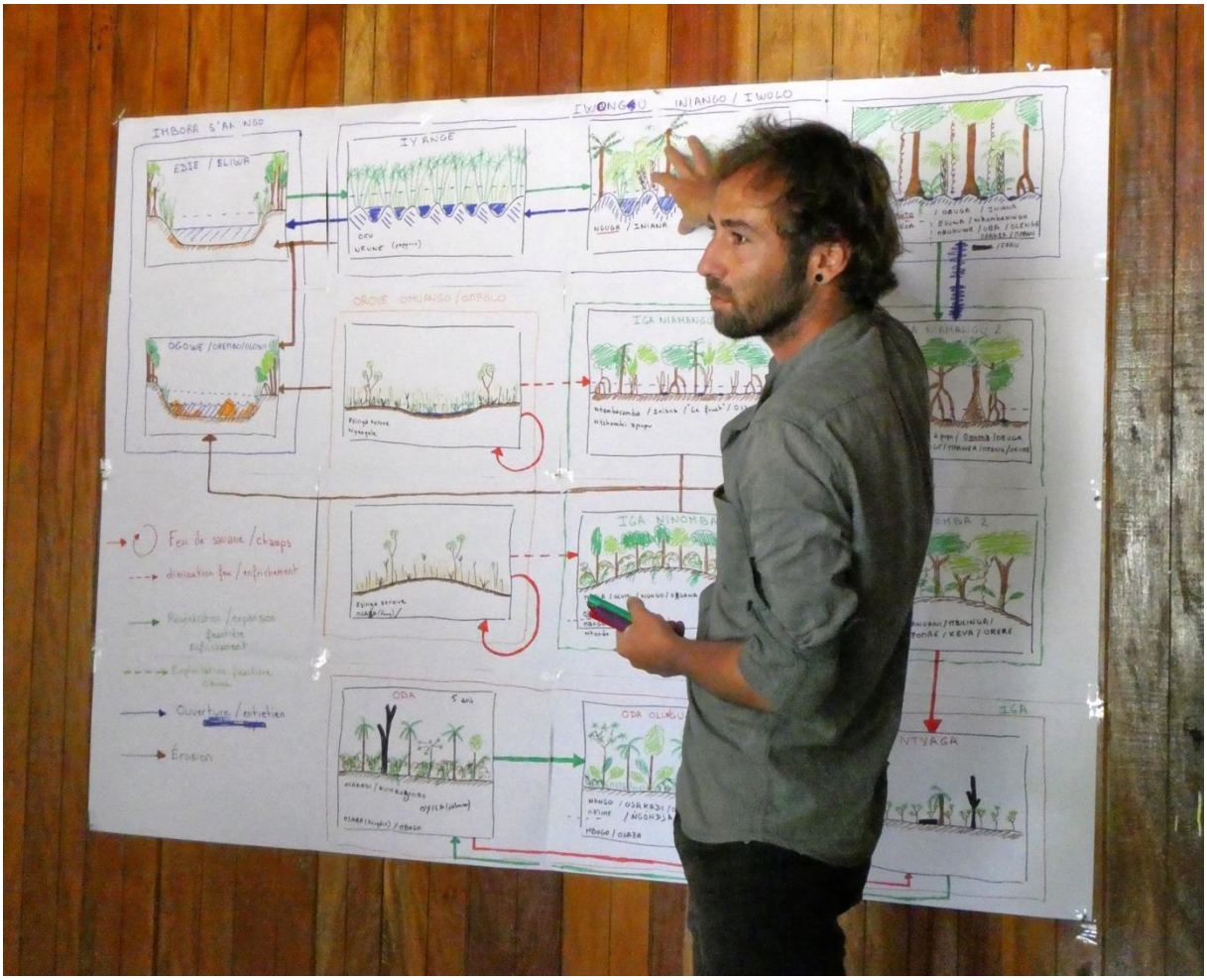

Photo 3: Discussion about landscape dynamics during the restitution workshop (N'Tyatanga, September 5, 2022).

4. DISCUSSION

4.1. Complementarity of local knowledge and remote sensing to develop a SES approach to the landscape

While wetland landscapes have long been difficult to map and analyze, notably because of spectral confusion with other land-cover classes and among different types of wetlands (Ozesmi & Bauer, 2002), technical advances in remote sensing now allow discrimination of these landscape classes. However, the methods used to study these landscapes around the world are often based on the production of indices and the use of complex machine-learning algorithms, such as *random forest*, that mostly require a series of satellite images to take into account the temporal variability in order to refine the statistical model (Adam et al., 2014; Mallick et al., 2021; Melton et al., 2022).

However, this type of method is more limited in the tropics. For example, in the present case, only one usable image was available in open access due to near-permanent annual cloud cover, making the use of *random forest* impossible. However, by using the high spatial resolution bands (10-m / 20-m) of a Sentinel-2 satellite image and combining the obtained information with a large quantity of field surveys (610 GPS points), proportionally to the area considered, we managed to produce a high-quality land-cover map. This map differentiates not only the three main terrestrial structures of the landscape as considered by the local population, i.e., swamps [Iwongu], forests [Iga] and savannahs [Orove], but also their different stages in terms of ecological succession, allowing not only the structure, but also the dynamics of the landscape to be highlighted. Thus, in areas where satellite imagery is still not widely available, field surveys are essential, and the quality of these surveys is determinant for the production of maps with a classification usable by remote sensing. In this study, the correspondence for all GPS points between field classification and their associated pixel values was highly significant (p-value < 0.001***), ensuring the production of a high-quality map without the necessity of a verification process.

However, the production of this map alone does not allow study of the landscape in its SES dimension. It is the collection of local knowledge that has allowed the development of a better understanding of the dynamics at work locally between the landscape and human populations. Indeed, long-standing studies suggest that inhabitants are in the best position to recognize and evaluate the changing ecological characteristics of the local landscape over time (Berkes et al., 2008; Azzurro, Moschella, & Maynou, 2011; Beaudreau & Levin, 2014). In this sense, their importance in the fight against climate change and the degradation of ecosystems requires no further demonstration (Mistry & Berardi, 2016). In our case, combining remote sensing analyses with local knowledge collected from the qualitative surveys allowed the development of an SES approach to the landscape, highlighting ecosystem services provided to local people by different components of the landscape through the analysis of their activities over time and space. This SES approach integrates the land-cover map with local practices and provides an essential tool for Ramsar site managers to identify the portions of the landscape that are most used by the population, in a way to implement integrated and locally acceptable conservation policies in the future. This SES approach also confirms similar studies conducted elsewhere in Africa,

demonstrating the contributions of local knowledge about human practices in landscape analyses using remote sensing (Yiran et al., 2012; Sulieman & Ahmed, 2013).

Finally, it is also important to note that in this study, local knowledge, not remote sensing, was used as the primary source of data in the field. Deploying an ethnographic approach to the initial analysis of the landscape allows looking at land use through the prism of perceptions and practices of local people. It is this understanding of the human and social dimensions of the SES that allowed definition of the landscape classes to be taken into consideration for remote-sensing analyses. The SES being a hybrid concept, taking into consideration the land uses before the biophysical environment allows a better integration of anthropic factors in efforts to apprehend the landscape as a complex system whose dynamics are governed by interactions between humans and environment. This point is particularly essential to ensuring the success of conservation programs for which the integration of local populations is an imperative prerequisite.

4.2. Local populations as drivers and guardians of landscape dynamics and changes over time

The landscape dynamics model we developed using an SES approach shows that, in the absence of human activities, the environment will gradually become more closed, with the disappearance of open ecosystems dominated by herbaceous vegetation in favor of forests in the long term. This dynamic of forest expansion at the expense of savannahs has been observed elsewhere in Gabon (Sebag et al., 2016) and in other countries in Central Africa (Demichelis et al., 2020; Ploton et al., 2022). In a context of climate change, these dynamics are expected to accelerate in African savannas in the coming decades (Stevens et al., 2016). Although the afforestation of these herbaceous areas may enhance carbon storage (although this point is debated [Zhou et al., 2022]), it is accompanied by a loss of biodiversity (Abreu et al., 2017). In fact, these savannah areas present a rich biodiversity, including species endemic to them, as in the savannahs of the Batéké Plateau National Park (Walters et al., 2022), and in other ancient savannahs around the world (Azevedo et al., 2016; Williams et al., 2017; Bond, 2019; Nerlekar et al., 2022). For this reason, the question of their conservation arises,

and recent opinions favor the protection of these landscapes and their associated biodiversity (Bond, 2019; Buisson et al., 2022).

As with most savannahs and herbaceous ecosystems around the world, fire and large mammalian herbivores are the two main obstacles to forest expansion (Bond, 2005). But with the reduction or extinction of large mammal populations, fires are now the dominant factor in places without livestock, and therefore the main driver of their equilibrium (Sebag et al., 2016; Demichelis et al., 2020; Ploton et al., 2022; Puttick et al., 2022). Locally, these fires are mainly set by hunters to attract wildlife from the forest to feed on the young regrowth of grasses. Thus, locally, human populations appear, through their practices, to be the main guardians of savannah areas and their associated biodiversity (in addition to the forest elephant and buffalo populations that are still present locally). This is especially true since demographic pressure is very low and deforestation is almost non-existent, with only 0.3% of the territory considered as deforested. The existence of ancient raised fields on the edge of floodable savannahs also testifies to the long co-evolutionary history between the local landscape and human populations. The departure of the populations would represent a hindrance to the conservation of biodiversity, and would require the implementation of fire management by the competent authorities. However, this approach to fire management shows its limits, as was the case in Lopé National Park, where despite fire management by park authorities, the surface area of the savannahs decreased after the prohibition of human activities within the park (Sebag et al., 2016).

Finally, the effects of local practices on the environment may not be positive in all components of the landscape. As the SES model shows, swampy areas are progressively colonized by forest, and this closure of the environment is accompanied by a progressive elevation of the soils and hence shortening of the flooding period, owing to an accumulation of organic matter. These wetlands are of primary interest for human populations, principally for fishing in the rainy season. However, they are also important in mitigating climate change. Peatlands are recognized as the largest terrestrial carbon stock per unit area (Leifeld & Menichetti, 2018), and a recent study has shown that *Cyperus papyrus* swamps, which constitute herbaceous swamps in our study area, stock more carbon than other types of herbaceous swamps in Central Africa (Kono et al., 2020). While fishing in the rainy season does not directly impact

these environments because the water level is high, certain practices contribute to the opening up of these environments near the villages, on the periphery of the lakes and along the Ogooué River. Indeed, in order to fish with machetes at the end of the dry season, or to create small ponds that attract fish during the rainy season, these areas can be cleared. Such clearing could lead to the degradation of these areas and the release of labile carbon into the river system, and of greenhouse gases into the atmosphere (Sonwa et al., 2020). However, this consideration must be tempered by the fact that the local population decline has resulted in a decrease in these practices, which are now very localized in areas that have been open for a long time. Climate change may be a greater threat to these areas than their clearing by humans. The increases in rainfall and in water levels observed locally by the population could also lead to the degradation of these areas throughout the Ramsar site, and therefore to release of carbon into the water and the atmosphere, as explained by Leng et al. (2019). For this reason, it seems essential to pay special attention to these specific landscape classes in terms of management and conservation at the scale of the Bas-Ogooué Ramsar site.

5. CONCLUSION

While field operators rely on land-cover maps to implement their development and conservation policies, this paper demonstrates that exclusive reliance on these maps results in significant loss of information about dynamics at the local scale. Without abandoning land-cover maps, which are still of major interest for assessing the spatio-temporal evolution of landscapes, this study highlights the interest of crossing remote sensing analyses with field data from local knowledge in order to develop a SES approach. We have shown that this approach provided an accurate and rich overview of interactions taking place within the landscape we studied.

Taking local knowledge into account therefore allows a more refined analysis of the landscape than a simple land-cover map. Integration of local knowledge better reflects local realities and reveals how local people use their landscape and associated resources. Where people are heavily dependent on natural resources, such insight is of crucial importance for developing effective biodiversity protection that takes into account the expectations and practices of local people. The results therefore show that

integrating local knowledge on land use with ecological data can refine our collective understanding of spatial and temporal landscape dynamics.

Moreover, this SES approach can more effectively align conservation efforts with expectations and practices of local populations and thereby create more concrete local buy-in. Conservation actors should therefore more systematically integrate local people's practices and perceptions into their understanding of landscape to improve the success of their protection programs, particularly with a view to better adaptation of societies and the landscapes they inhabit to global change.

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